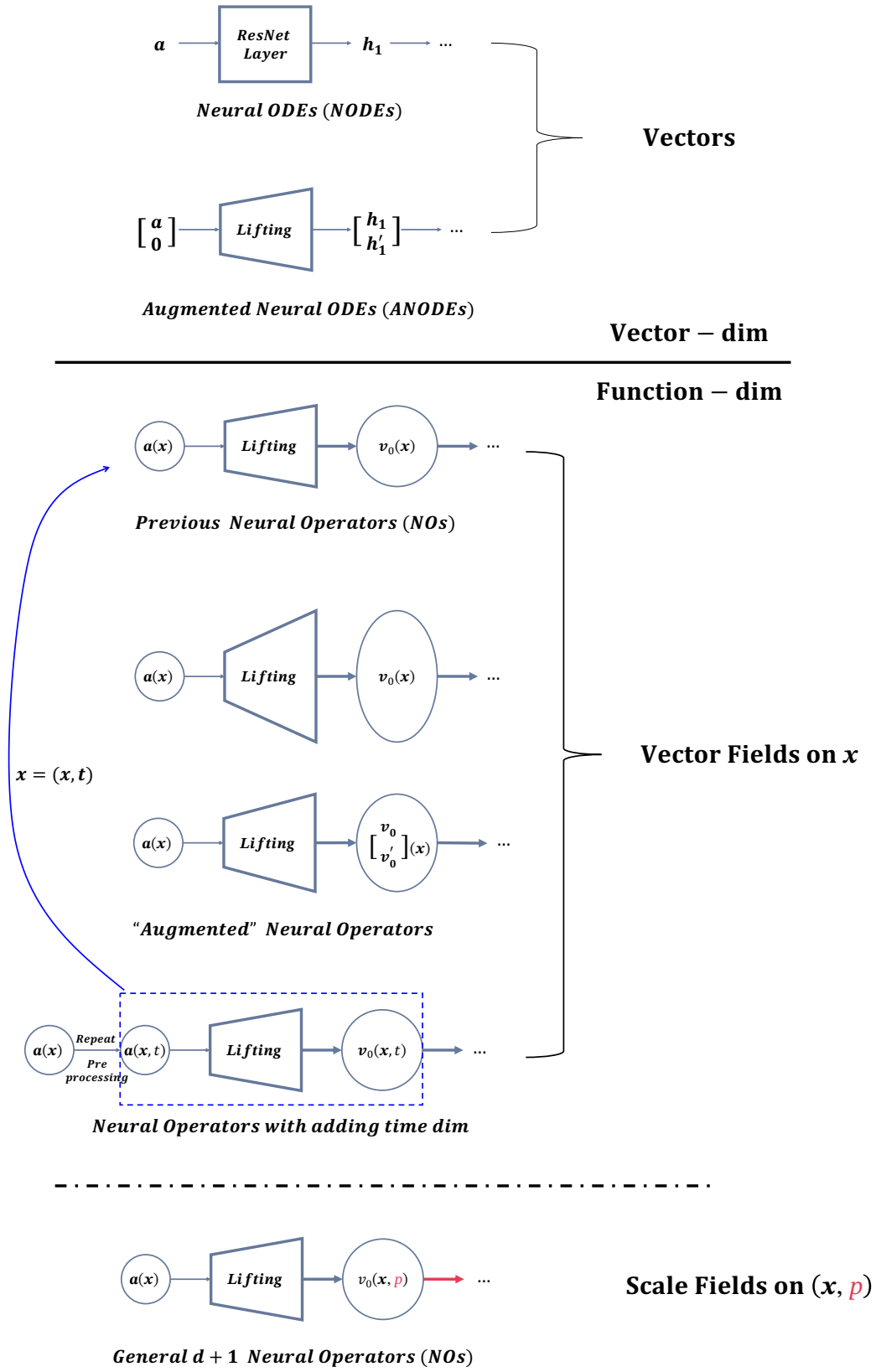

Figure 1. Architectural classifications by input and "dimension" type. **Top**: Neural ODEs (NODEs) and Augmented NODEs (ANODEs) operate on finite-dimensional vector spaces, using manual concatenation for lifting. **Middle**: Existing Neural Operators (NOs) propagate d dimensional vector fields, including variants that append a temporal dimension. **Bottom**: General $d+1$ Neural Operators utilize an auxiliary embedding dimension to propagte $(d+1)$ dimensional scalar fields. [Clarifies the differences in "lifting" among ANODEs, 3D-FNO (i.e., FNO with the spatial field manually repeated along the time dimension), and NOs defined in $d+1$ dimensions, in response to Reviewers fuFw (beginning part of the Weaknesses) / dr2G (W3).]

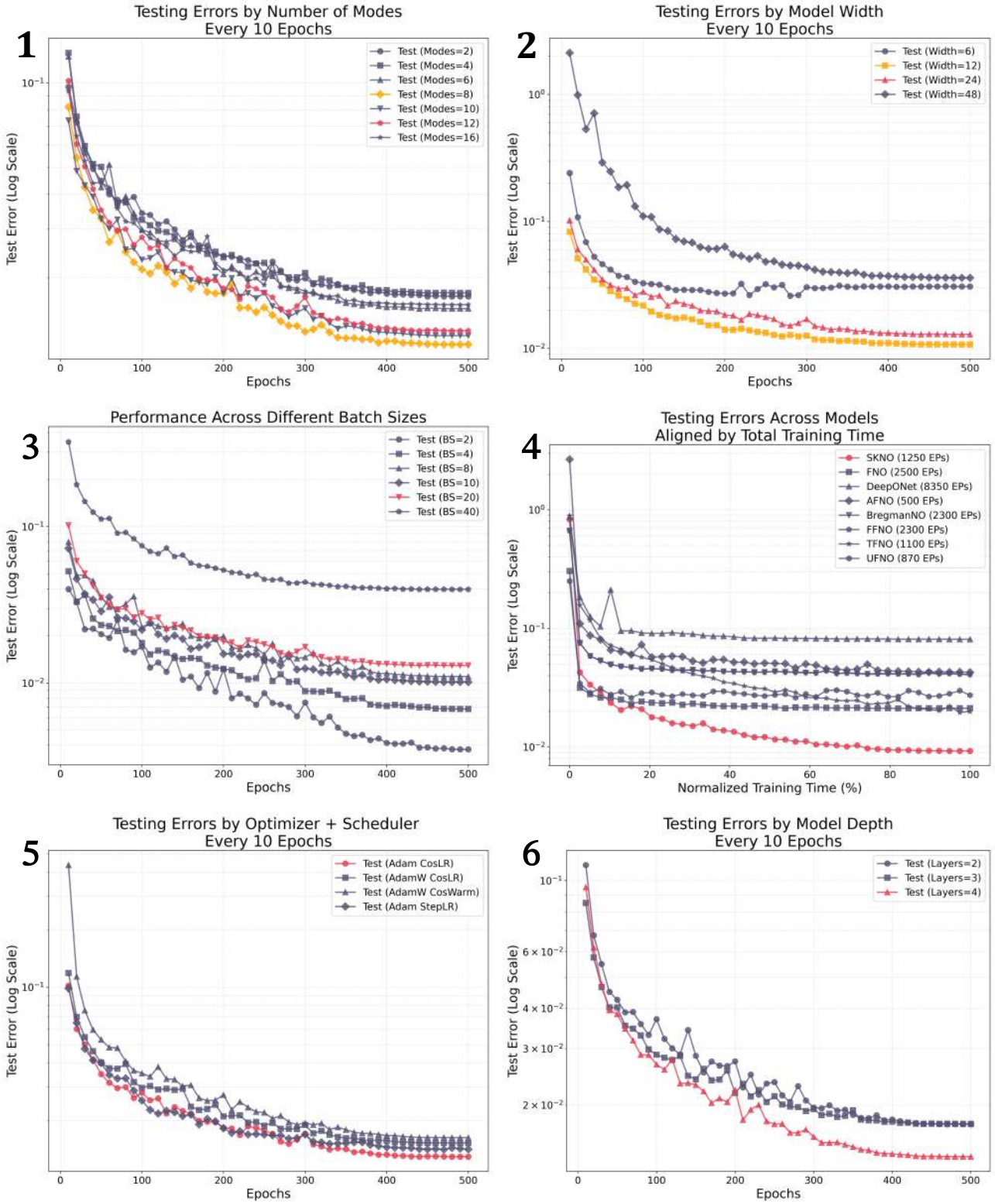

Figure 2. Ablation Studies and Performance Comparison of SKNO on the 2D Gray-Scott. All red lines represent the default configurations of SKNO used in baseline comparisons. **1** and **2**: These panels illustrate SKNO's performance under different truncation modes in the original domain and varying sizes along p-dim (model width). The yellow lines highlight that SKNO can achieve optimal performance with a smaller number of frequency truncation modes and a more compact p-dimension size. **3**, **5**, and **6**: These plots show the influence of batch size, choice of optimizer and learning rate scheduler, and network depth on the model's convergence and final test error. **4**: This panel displays the test error decay curves for various models aligned by the same total training time, where SKNO successfully surpasses others by above 55.79% error reduction (*Best testing performance: SKNO with* 9.245e-3 *and FNO with* 2.091e-2). [Addresses Reviewers fuFw (W5) / PLoT (W3, Q2) / dr2G (W4, Q5)]

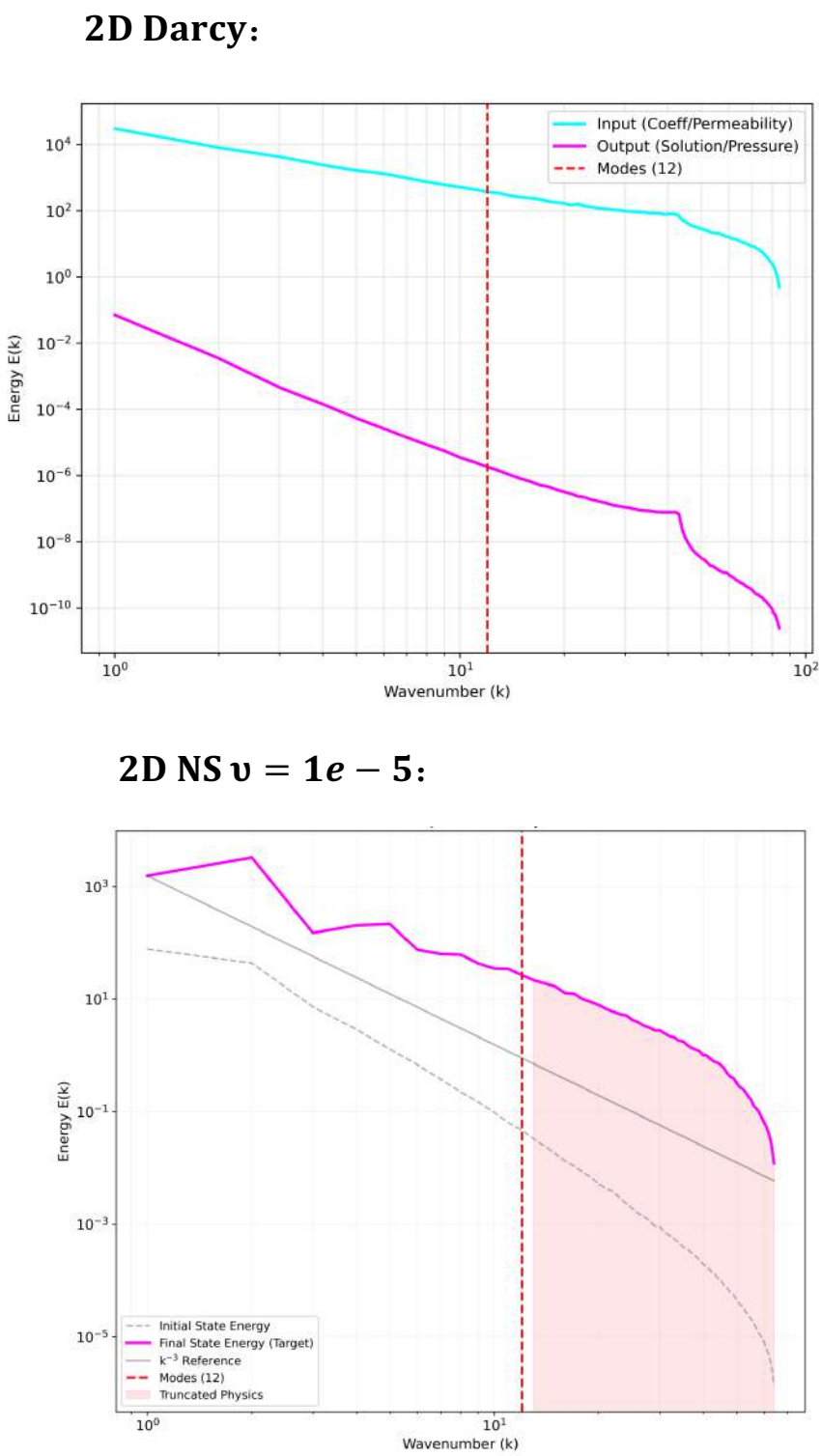

Figure 3. Energy spectral decay for the 2D Darcy flow and 2D Navier-Stokes datasets. The spectral distribution of the data provides insight into the complexity of the operator learning task. Notably, both datasets maintain substantial energy in high-frequency modes, spanning a broad and highly energetic frequency band (note that the wavenumber is plotted on a logarithmic scale). Furthermore, increased operator nonlinearity results in a more irregular spectral decay in the solution (magenta line). In both scenarios, the SKNO model demonstrates superior performance, achieving significantly lower error levels than competing models. [Addresses Reviewers fuFw (W2) / dr2G (W4)]

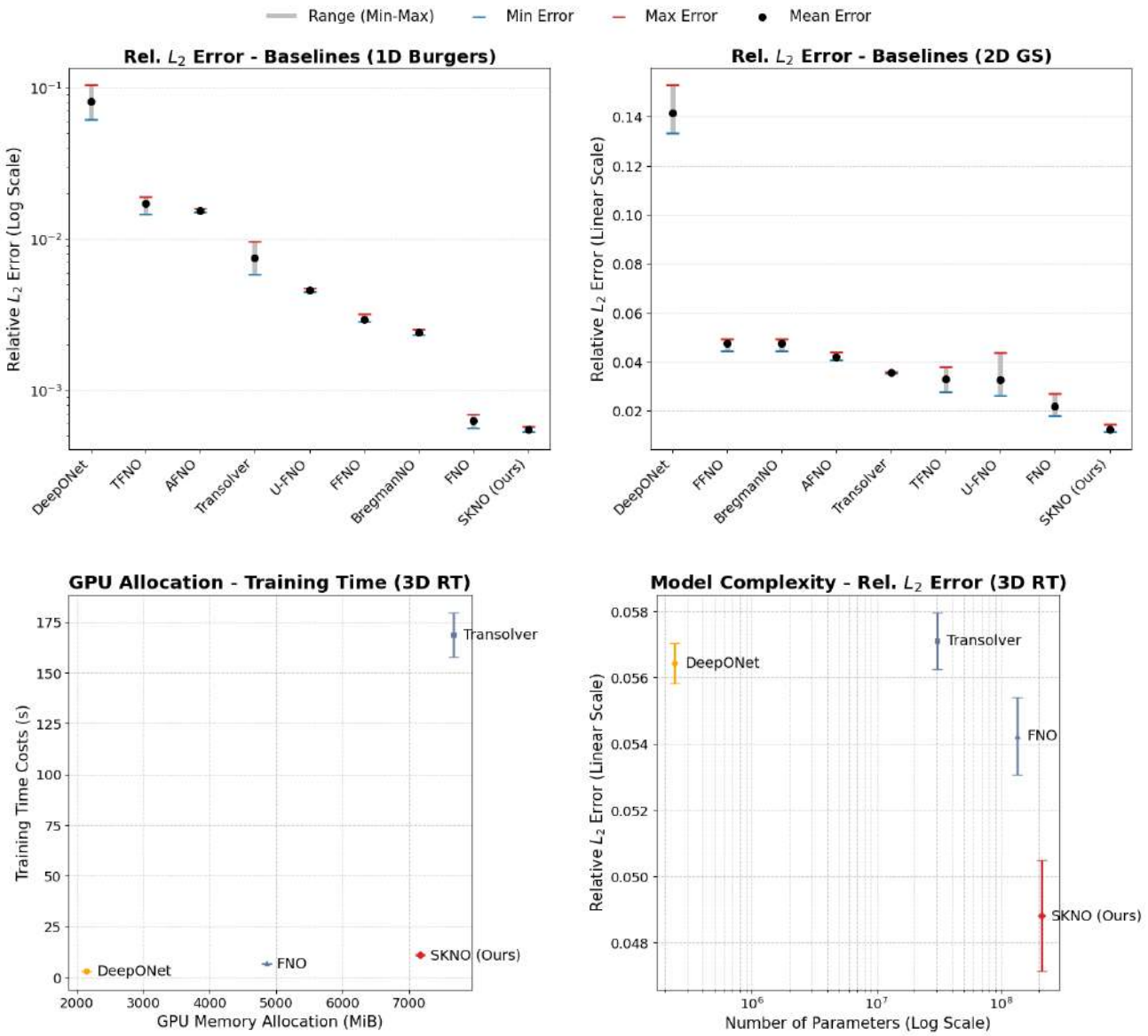

Figure 4. Visualizations of statistical performance across multiple runs with six random seeds. **Top Row**: On both the 1D Burgers and 2D Gray-Scott problems, our SKNO achieves the best performance within the same number of training epochs compared to all baselines. **Bottom Row**: On the 3D Rayleigh-Taylor Instability problem, while Transolver employs multi-head mechanisms to mitigate the explosive growth in training time caused by long embeddings, SKNO utilize in the transformed domain of the introduced auxiliary dimension, which drastically reduces training time while delivering a performance leap that remains unattainable for other baseline methods. [Addresses Reviewers fuFw (W1, W2, W4, W5, Q3) / PLoT (W1, W4) / dr2G (W4, Q3)]

Table 1. Performance comparison on the 2D Darcy flow problem evaluating alternative implementations within d dimensional and $(d+1)$ dimensional frameworks. The results are categorized into three distinct blocks. **Top:** Baseline d dimensional Neural Operators (NOs). **Middle:** Architectures utilizing lifting within the vector space. Specifically, *Augmented FNO* represents an FNO with expanded channel width $(+1)$. *Hyperbolic FNO* introduces an auxiliary vector-dimension modeled as a Lorentz hyperboloid, which creates a curved feature space with constant negative curvature to capture hierarchical scale levels. *2Stack-FNO* stacks two Fourier layers on the d-dimensional domain (parallel), omitting specialized operator design along the p-dimension. **Bottom:** *SKNO-Attention* replaces the p-dimensional spectral operator in SKNO with the attention-based aggregation module from Transolver, compared against our proposed SKNO. *(Adapted from Table 11 in lines 1365-1373 of our manuscript, with two "augmented" baseline methods added.)* [Addresses Reviewers fuFw (W4, Q2, and beginning part of the Weaknesses) / PLoT (W3) / dr2G (W4, Q5)]

Model	DeepONet	GNOT	OFormer	LSM	CNO	FNO	Transolver
Rel. L_2 Err.	0.0609	0.0102	0.0121	0.0065	0.0060	0.0062	0.0058
Training Time	**5.20 min**	22.58 min	21.91 min	20.37 min	63.34 min	10.21 min	115.48 min

Model	*Augmented FNO*	*Hyperbolic FNO*	*2Stack-FNO*
Rel. L_2 Err.	0.0067	0.0080	0.0061
Training Time	10.82 min	15.83 min	13.35 min

Model	*SKNO-Attention*	SKNO (Ours)
Rel. L_2 Err.	0.0058	**0.0055**
Training Time	76.90 min	17.59 min

Table 2. Comparison of training times and relative L_2 errors for various models evaluated across six independent random seeds. FNO (time dim)*: a variant specifically designed for spatio-temporal function propagation. Within the context of this study, this variant operates directly on d dimensional inputs rather than on the auxiliary embedding dimension after lifting. c.s.: coming soon. *Current results show good performance of SKNO across seeds. Due to the limited rebuttal time and computational resources (a single NVIDIA 4090), the results will be gradually updated during the whole rebuttal period.* [Addresses Reviewers fuFw (W1, W2, W4), PLoT (W1, W3, W4, Q2) and dr2G (W3, W4, Q3, Q5)]

Model	**Seed 0**		**Seed 1**		**Seed 2**	
	Training Time	Rel. L_2 Err.	Training Time	Rel. L_2 Err.	Training Time	Rel. L_2 Err.
DeepONet	874.50	5.414e-1	873.16	3.486e-1	885.19	3.534e-1
AFNO	1192.37	3.085e-1	1127.28	3.105e-1	c.s.	c.s.
TFNO	749.36	2.270e-1	769.06	2.245e-1	733.09	2.263e-1
U-FNO	981.50	1.976e-1	986.20	1.970e-1	999.03	1.949e-1
FFNO	754.98	2.172e-1	c.s.	c.s.	c.s.	c.s.
BregmanNO	754.87	2.111e-1	c.s.	c.s.	c.s.	c.s.
FNO	843.37	1.277e-1	856.40	1.285e-1	841.03	1.356e-1
FNO (time dim)*	787.14	1.642e-1	786.21	1.663e-1	803.59	1.618e-1
Transolver	1263.33	1.002e-1	1269.94	2.086e-1	1469.21	8.909e-2
SKNO (Ours)	**720.28**	**8.374e-2**	**706.68**	**8.726e-2**	**736.79**	**8.355e-2**

Model	**Seed 3**		**Seed 4**		**Seed 5**	
	Training Time	Rel. L_2 Err.	Training Time	Rel. L_2 Err.	Training Time	Rel. L_2 Err.
DeepONet	872.83	3.482e-1	878.85	3.444e-1	880.26	3.481e-1
AFNO	1117.37	3.062e-1	1110.54	3.052e-1	c.s.	c.s.
TFNO	770.26	2.314e-1	755.34	2.363e-1	768.82	2.355e-1
U-FNO	1002.90	2.000e-1	984.78	2.027e-1	997.70	1.917e-1
FFNO	c.s.	c.s.	c.s.	c.s.	c.s.	c.s.
BregmanNO	c.s.	c.s.	c.s.	c.s.	c.s.	c.s.
FNO	c.s.	c.s.	c.s.	c.s.	c.s.	c.s.
FNO (time dim)*	789.53	1.630e-1	784.37	1.632e-1	796.33	1.655e-1
Transolver	1384.41	8.406e-2	1285.63	1.008e-1	c.s.	c.s.
SKNO (Ours)	**708.93**	**8.379e-2**	**738.20**	**8.460e-2**	720.98	**8.077e-2**

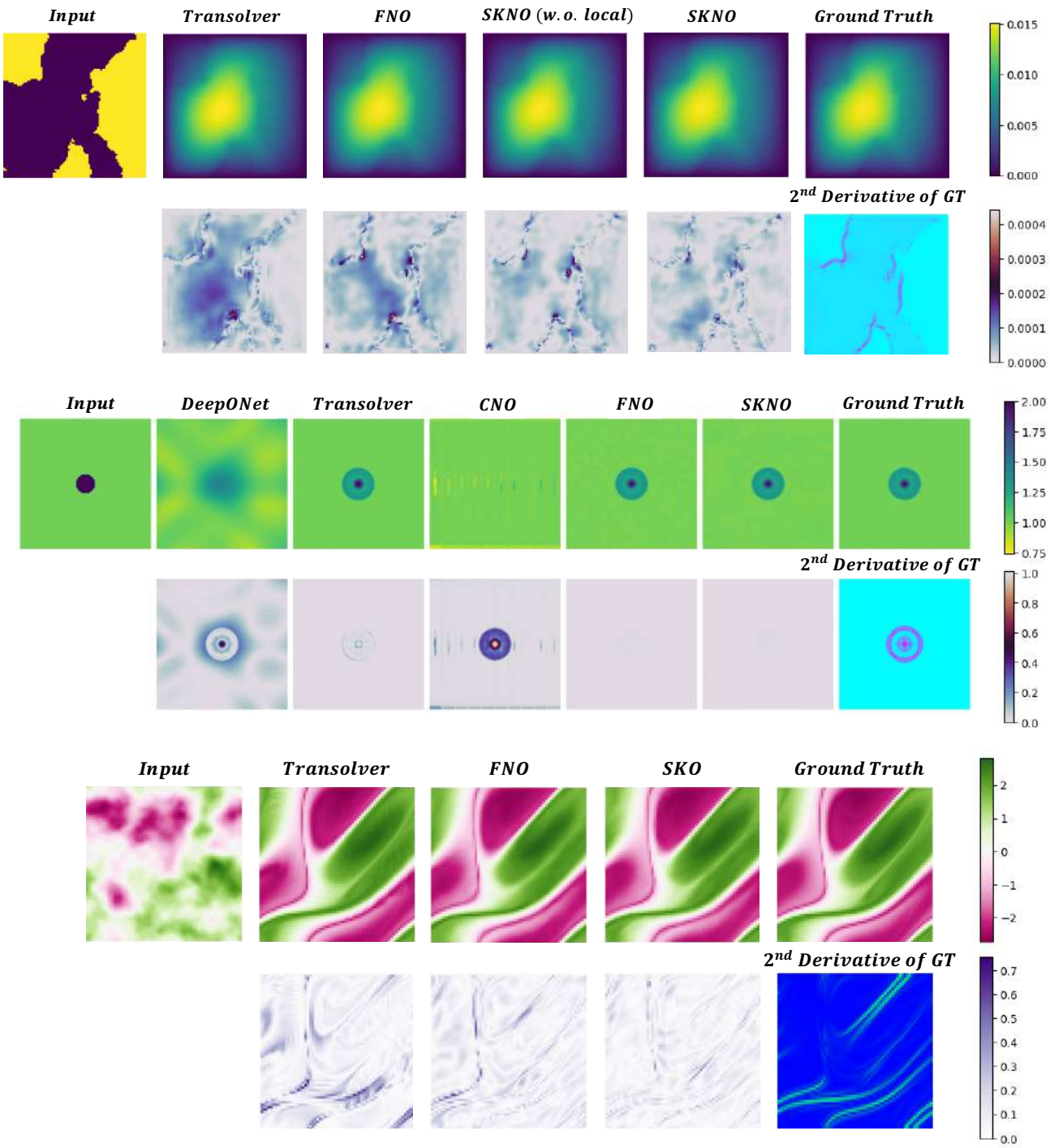

Figure 5. Visualization of predictions, errors, and the second derivative of the ground truth for 2D Darcy flow, 2D Shallow-Water equations (using randomly mixed training data), and 2D Navier-Stokes (NS) equations. In each block, the top row displays the input field, predictions from various baseline models alongside our proposed method (SKNO), and the ground truth. The bottom row presents the corresponding error visualizations for each model's prediction, as well as the second derivative of the ground truth. Overall, these comparisons illustrate the consistently low error of SKNO on physical field mapping operators across both microscopic and macroscopic scales. *(Combined and adapted from Figures 10, 11, and 14 in our original manuscript [lines 1462-1482, 1485-1504, and 1578-1594], with color bars added.)* [Addresses Reviewer fuFw (Q9)]

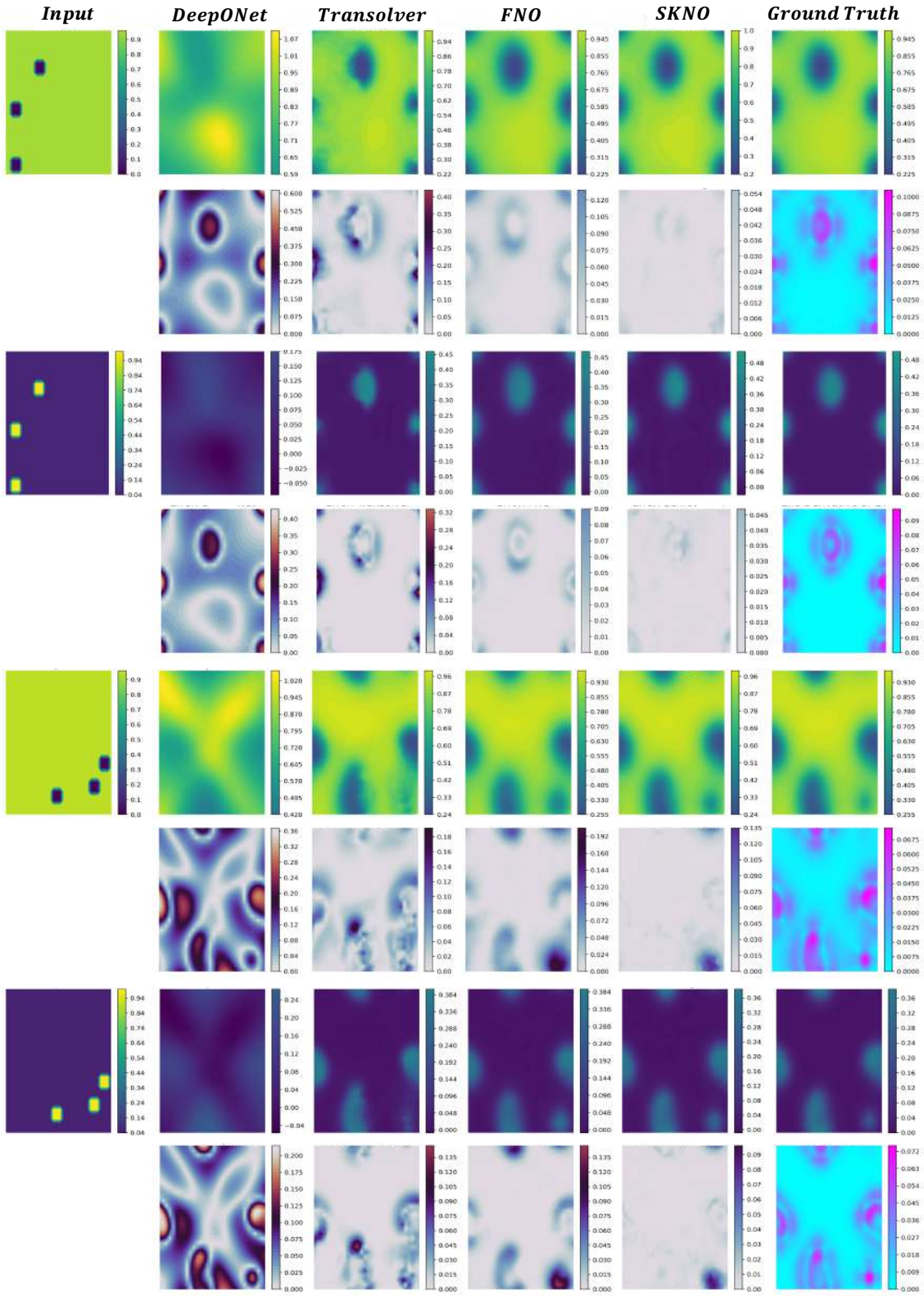

Figure 6. Visualization of model predictions for the 2D Gray-Scott dataset. The top row of each block displays the field $u(x)$, while the bottom row displays the field $v(x)$. Across all presented cases and physical fields, SKNO significantly suppresses error artifacts and reconstructs the ground truth with vastly superior precision compared to the baseline architectures. *(Adapted from Figure 13 in lines 1540-1577 of our manuscript, with one more showcase and corresponding color bars added.)* [Addresses Reviewer fuFw (Q9)]

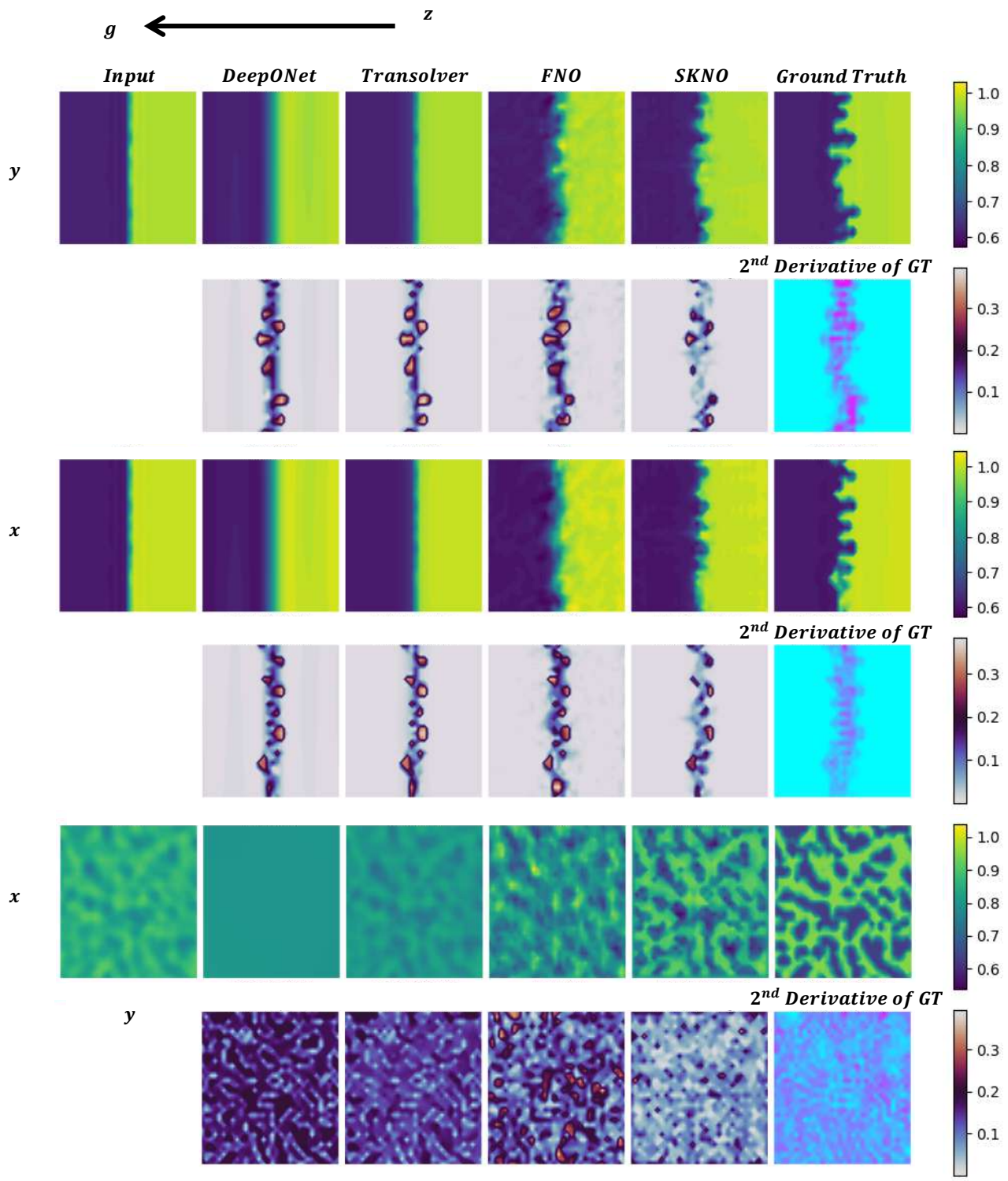

Figure 7. Visualization of the 3D Rayleigh–Taylor instability case. All slices are parallel to the coordinate planes, and their intersection points are chosen as the grid points nearest to the center of the box. We show a representative extreme case, namely predicting the output field after 20 steps from a perturbed initial condition. While other lightweight methods largely fail, SKNO can still capture the overall flow trends and the interfacial surfaces between the fluids with different densities. $\boldsymbol{g}$ represents the gravity direction. *(Adapted from Figure 15 in Page 30 of our manuscript, with color bars added.)* [Addresses Reviewer fuFw (Q9)]

Table 3. Math Symbol and Descriptions [Supports the rebuttal to Reviewers PLoT (W2, Q1) / 7b5C (W1, Q1) / dr2G (W2, Q2)]

Math Symbol	Description
N_p	p dim size / channel width
$\Phi(\cdot)$ or $F(\cdot)$	p-domain aggregation transform
$\Phi^{-1}(\cdot)$ or $F^{-1}(\cdot)$	p-domain inverse transform
V_k	Branch A linear weights (per spatial mode)
W_k	Branch B linear weights (per spatial mode)
$\sigma_\Phi(M_2\sigma_\Phi(M_1\cdot))$	Nonlinear MLP in Φ-domain
$\sigma(\cdot)$	Nonlinear activations
Φ	Invertible transform along p-dim
F	Discretized Fourier Transform, as a special case of Φ
$\mu(\Phi)$	$\max_{i\neq j} \lvert\langle\phi_i,\phi_j\rangle\rvert/(\lVert\phi_i\rVert_2\lVert\phi_j\rVert_2)$
$\mathcal{T}_1$	$\{\sigma(W\cdot) : W \in \mathbb{C}^{C\times C}\}$
$\mathcal{T}_2$	$\{\Phi^{-1}M_2\sigma(M_1\Phi\cdot)\}$
$\mathcal{T}_3$	$\{\sigma(\Phi^{-1}V\Phi\cdot)\}$
L	The number of layers
N_x	The number of grid points in d dimension / domain D_x

Table 4. Summary of benchmark dataset characteristics for the evaluated problems. The table details the number of samples in the training and testing sets, the spatial resolution or number of grid points (N_x), the input feature dimensionality (d_a), and the output field dimensionality (d_u). *(As originally detailed in lines 683–809 of our manuscript.)* [Addresses Reviewer fuFw (W2, Q6)]

BENCHMARK	TRAIN. SAMPLES	TEST. SAMPLES	N_x	d_a	d_u
1D HEAT (TIME)	1000	100	256	1	1
1D ADVECTION (TIME)	1000	200	256	10	1
1D BURGERS (TIME)	1000	100	128	1	1
2D STRESS	1000	100	2304	1	3
2D STRAIN	1000	100	2304	1	3
2D DARCY	1000	100	7225	1	1
2D GRAY-SCOTT (TIME)	200	20	1024	20	2
2D SHALLOW WATER (TIME)	500	300	1024/4096/16384[†]	10	1
2D INCOMPR. NAVIER-STOKES (1E-4, TIME)	1000	200	4096	10	1
2D INCOM. NAVIER-STOKES (1E-5, TIME)	1000	200	4096	10	1
3D COM. NAVIER-STOKES (TIME)	500	100	32768	15	5
3D RAYLEIGH-TAYLOR INSTAB. (TIME)	220	40	32768	4	4
ERA5 WIND FIELD PREDICTION (TIME)	64	16	262144	4	1

[†]For the 2D Shallow Water benchmark, spatial resolutions (N_x) are randomly downsampled to grids of 32×32, 64×64, and 128×128 (maximum).

Table 5. Comparison of parameter counts across different models and problems. [Addresses Reviewers fuFw (W5, Q3) / dr2G (W4, Q5)]

Problem	DeepONet	Transolver	FNO	SKNO
2D Gray-Scott	86,802	11,237,698	1,337,450	2,141,844
2D Shallow-Water	16,844,624	2,175,473	599,833	971,636
2D Navier-Stokes	4,261,712	11,232,321	1,337,113	2,149,884
3D Navier-Stokes	249,989	30,339,653	134,235,877	210,028,292
3D RT	240,420	30,333,764	134,235,396	210,061,700

Table 6. Capacity Efficiency on 2D Gray-Scott. Comparison of SKNO against FNO under systematically varied embedding sizes ($1\times$ to $4\times$), parallel stacking (2/3/4-Stack), and branch architecture (B+B vs. A+B), evaluated over 6 random seeds with 500 training epochs. SKNO (A+B) achieves the lowest error while using fewer parameters and fewer FLOPs than FNO at any tested scaling configuration. FNO scaling is non-monotonic, peaking at $2\times$ embedding size and degrading beyond. SKNO(B+B), which replaces the Fourier-domain branch along p-dim with a duplicate standard branch at matched parameter budget, fails to improve over FNO, confirming the architectural origin of SKNO's gains. [Addresses Reviewers PLoT (W3, Q2) / dr2G (W4, Q5)]

Model	**# Params**	**Rel. L_2 Err. $\pm$ Std.**	**FLOPs**
		FNO with varying embedding size	
FNO	1,337,450	2.169e-2 $\pm$ 3.35e-3	940,318,720
FNO (+1 emb.)	1,451,177	2.042e-2 $\pm$ 1.24e-3	998,184,960
FNO (1.33$\times$ emb.)	2,377,186	1.869e-2 $\pm$ 2.21e-3	1,445,109,760
FNO (1.5$\times$ emb.)	3,008,414	1.834e-2 $\pm$ 2.67e-3	1,733,386,240
FNO (2$\times$ emb.)	5,347,538	1.685e-2 $\pm$ 3.73e-3	2,741,739,520
FNO (3$\times$ emb.)	12,030,266	1.952e-2 $\pm$ 1.83e-3	5,404,303,360
FNO (4$\times$ emb.)	21,385,634	1.839e-2 $\pm$ 1.10e-3	8,928,010,240
		Parallel stacking in the same basis	
2-Stack FNO	2,685,962	2.532e-2 $\pm$ 4.29e-3	2,328,371,200
3-Stack FNO	4,027,370	2.214e-2 $\pm$ 2.26e-3	3,429,376,000
4-Stack FNO	5,368,778	2.063e-2 $\pm$ 3.30e-3	4,530,380,800
		SKNO branch ablation	
SKNO (B+B)	2,690,762	2.447e-2 $\pm$ 9.65e-4	2,521,047,040
SKNO (A+B)	**2,141,844**	**1.256e-2 $\pm$ 1.14e-3**	**2,112,141,236**